SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High) Assessment
Tool Weightage: 40%

QUESTIONS		
Q. No	**Question**	**Bloom's Level**
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

Code of Conduct:

I A.Rakesh (Reg No: 192525388) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1.Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

Schema and Instance

Schema

A **schema** is the overall design or structure of a database. It defines how data is organized, including tables, attributes, relationships, and constraints. A schema is relatively stable and changes rarely.

This structure (table name and column names) is the schema.

Instance

An **instance** is the actual data stored in the database at a particular moment. It changes frequently as records are added, updated, or deleted.

Example:

Roll_No	Name	Department	Age
101	Ravi	CSE	20
102	Priya	ECE	19

These records represent the current instance of the database.

Difference Between Schema and Instance

Schema	Instance
Structure or blueprint of the database	Actual data stored in the database
Changes rarely	Changes frequently
Defined during database design	Changes with insert, update, and delete operations
Also called **intension**	Also called **extension**

Logical Schema, Physical Schema, and View Schema

Logical Schema	Physical Schema	View Schema
Describes the logical structure of the entire database.	Describes how data is physically stored in memory or disk.	Describes only the part of the database visible to a particular user.
Includes tables, attributes, relationships, and constraints.	Includes storage methods, indexes, file organization, and access paths.	Includes customized views for different users.

Independent of physical storage.	Dependent on storage devices and implementation.	Provides security by hiding unnecessary data.

Example

Suppose there is an **Employee** database.

1. Logical Schema

Employee(Emp_ID, Name, Department, Salary)

This defines the database structure.

2. Physical Schema

- Employee records stored in data files.
- Index created on Emp_ID for faster searching.
- Data stored using B+ tree indexing. **3. View Schema**

Manager View:

Emp_ID, Name, Department, Salary

Employee View:

Emp_ID, Name, Department

Employees cannot see salary details, while managers can.

2.Bank Management System – Entities, Attributes, Relationships, and ER Diagram

1. Entities and Attributes

1. Customer

- **Customer_ID** (Primary Key)
- Name
- Address
- Phone_No

- Email
- DOB

2. Account

- **Account_No** (Primary Key)
- Account_Type • Balance
- Opening_Date
- Customer_ID (Foreign Key)

3. Branch

- **Branch_ID** (Primary Key)
- Branch_Name
- Location
- IFSC_Code

4. Employee

- **Employee_ID** (Primary Key)
- Name
- Designation
- Salary
- Branch_ID (Foreign Key)

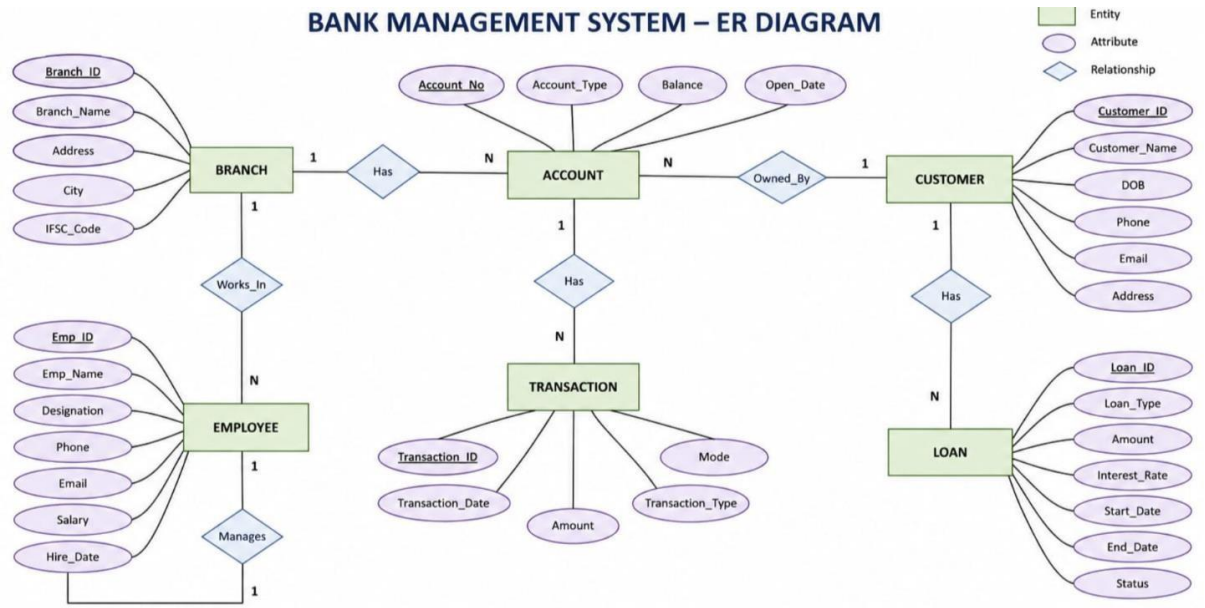
5. Loan

- **Loan_ID** (Primary Key)
- Loan_Type
- Loan_Amount
- Interest_Rate
- Customer_ID (Foreign Key)

2. Relationships

Relationship	Description	Cardinality
Customer – Account	A customer can have one or more accounts.	**1 : M**
Branch – Account	A branch manages many accounts.	**1 : M**
Branch – Employee	A branch employs many employees.	**1 : M**
Customer – Loan	A customer can take multiple loans.	**1 : M**

3. ER Diagram (Text Representation)



3. Explain the Extended ER (EER) Model. Describe the concepts of Generalization, Specialization, and Aggregation.

Extended Entity-Relationship (EER) Model

The **Extended Entity-Relationship (EER) Model** is an advanced version of the **EntityRelationship (ER) Model** used in database design. It extends the basic ER model by introducing additional concepts such as **generalization, specialization, aggregation, inheritance, categories (union types), and constraints**. These features help represent complex real-world situations more accurately and efficiently.

The EER model is widely used in designing large and complex database systems because it supports hierarchical relationships and reduces redundancy.

Features of the EER Model

- Supports **inheritance** of attributes and relationships.
- Represents **superclass–subclass** relationships.
- Reduces **data redundancy**.
- Models **complex database structures** effectively.
- Improves database organization and flexibility.
- Provides better semantic representation of real-world applications.

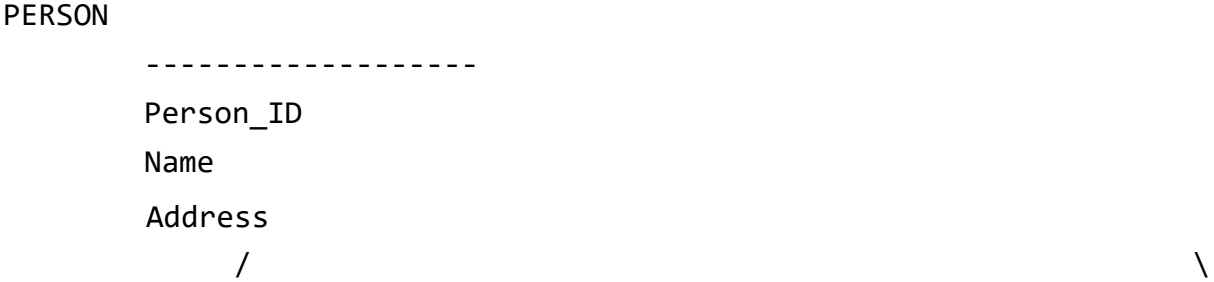
1. Generalization

Definition

Generalization is the process of combining two or more lower-level entities that have common attributes into a single higher-level entity called a **Superclass**.

It follows a **bottom-up approach**, where similar entities are merged into a generalized entity.

Example



Explanation

- **Person** is the **Superclass**.
- **Student** and **Employee** are **Subclasses**.
- Common attributes such as **Person_ID, Name, and Address** are stored only in the superclass.
- The subclasses inherit these common attributes and also contain their own specific attributes.

Advantages

- Eliminates duplicate data.
- Reduces redundancy.
- Simplifies database design. • Promotes attribute inheritance.

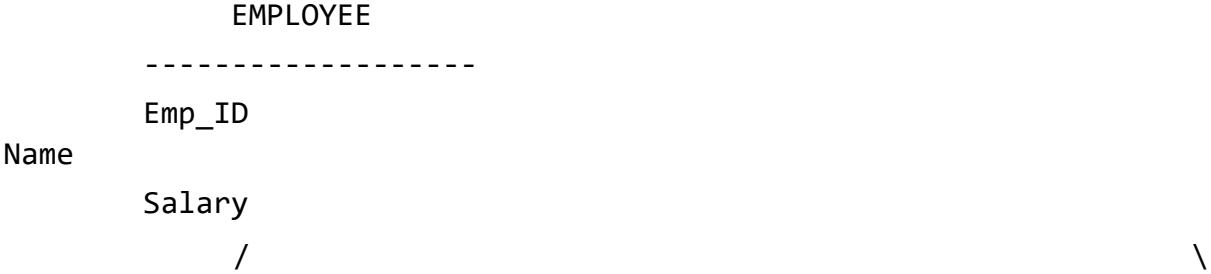
2. Specialization

Definition

Specialization is the process of dividing a higher-level entity (Superclass) into two or more lower-level entities (Subclasses) based on distinguishing characteristics.

It follows a **top-down approach**, where a general entity is divided into more specific entities.

Example



Explanation

- **Employee** is the **Superclass**.
- **Manager** and **Engineer** are **Subclasses**.
- All employees share common attributes such as **Emp_ID, Name, and Salary**.
- Managers have an additional attribute **Bonus**.
- Engineers have an additional attribute **Skill**.

Advantages

- Represents specialized categories.
- Improves data organization.
- Makes database design more flexible. • Allows subclass-specific attributes and relationships.

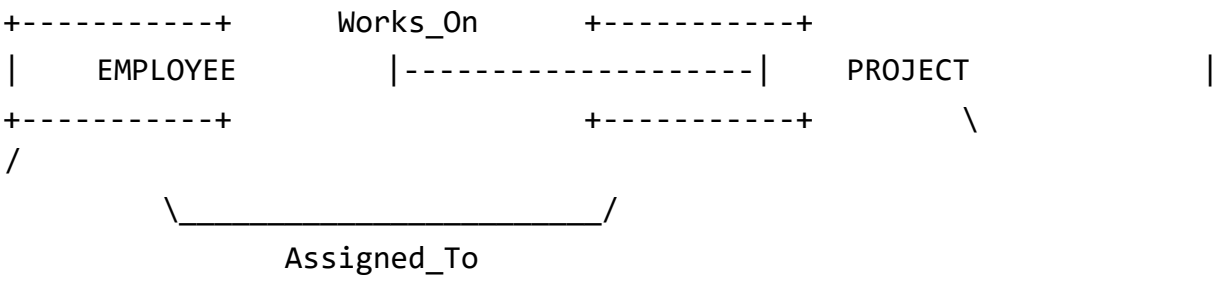
3. Aggregation

Definition

Aggregation is a technique used to represent a relationship between a relationship and another entity. It treats a relationship set as a higher-level entity so that it can participate in another relationship.

Aggregation is useful when a relationship itself needs to be associated with another entity.

Example



Explanation

- An **Employee** works on a **Project** through the **Works_On** relationship.
- Instead of linking **Department** separately to Employee or Project, the **Works_On** relationship is aggregated.
- The **Department** is associated with the entire **Works_On** relationship.
- Thus, aggregation allows relationships to participate in other relationships.

Advantages

- Represents complex real-world situations.
- Improves semantic clarity.
- Avoids unnecessary entities. • Makes database modeling more accurate.

Differences Between Generalization, Specialization, and Aggregation

Generalization	Specialization	Aggregation
Combines lower-level entities into a superclass.	Divides a superclass into subclasses.	Treats a relationship as a higher-level entity.
Bottom-up approach.	Top-down approach.	Models relationships between relationships.
Focuses on identifying common features.	Focuses on identifying unique features.	Focuses on complex relationship representation.
Reduces redundancy.	Adds detailed classification.	Represents higher-level relationships.
Promotes inheritance.	Allows subclassspecific attributes.	Improves semantic representation of relationships.

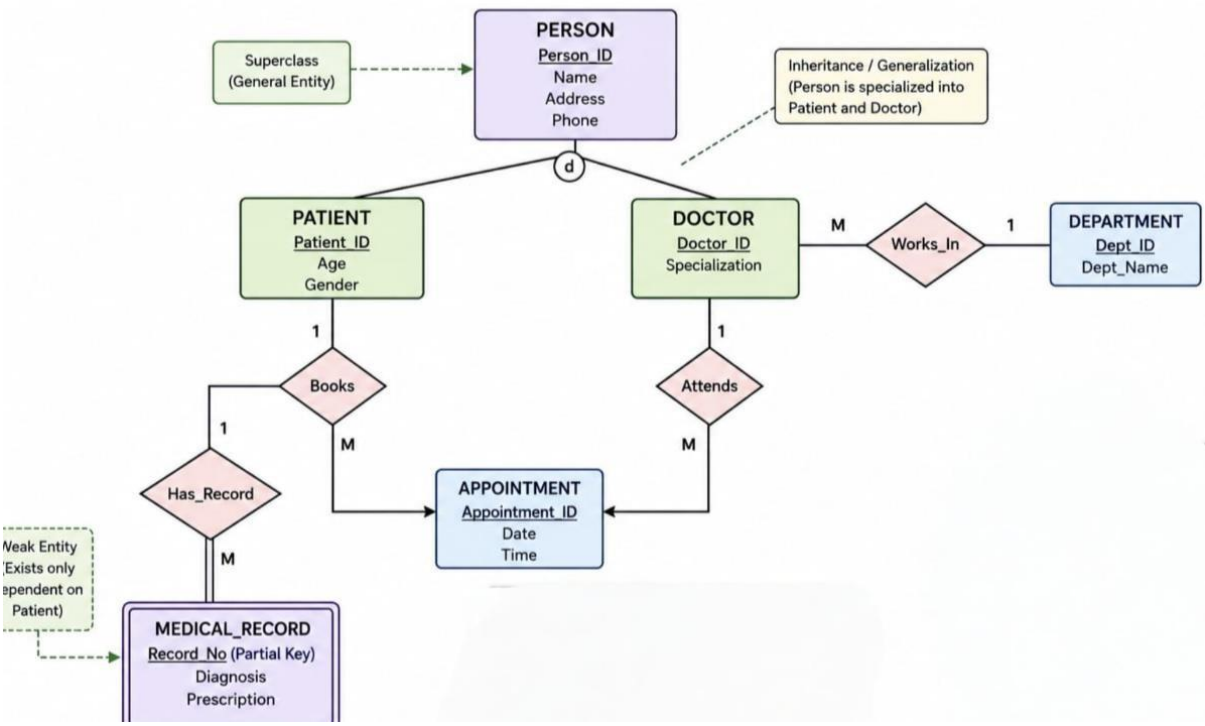
Advantages of the EER Model

- Represents complex real-world applications effectively.
- Supports inheritance and hierarchical relationships.

- Reduces data redundancy.
- Improves database consistency.
- Makes database design easier to understand and maintain. • Supports object-oriented concepts in database design.

4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

EER Diagram:



Explanation:The Hospital Management System EER diagram consists of Person as the superclass, with Patient and Doctor as its subclasses, demonstrating inheritance (specialization). The subclasses inherit common attributes such as Person_ID, Name, Address, Phone, and Date of Birth, while each has its own specific attributes. Other entities include Ward and Nurse, with relationships such as Treats (Doctor–Patient), Admitted_To (Patient–Ward), and Assigned_To (Nurse–Ward), which model the hospital's daily operations.

The diagram also includes Medical_Record as a weak entity, since it cannot exist without a Patient. It is connected through an identifying relationship (Has) and is identified using the combination of Patient_ID and Record_No (partial key). This EER model reduces data redundancy through inheritance, accurately

represents real-world hospital relationships, and efficiently manages patient, doctor, ward, nurse, and medical record information.

5.Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Role of a Database Administrator (DBA)

A **Database Administrator (DBA)** is a professional responsible for managing, maintaining, and securing an organization's database systems. The DBA ensures that databases operate efficiently, remain secure, and are always available to authorized users. They are responsible for installing database software, monitoring performance, performing backups and recovery, controlling user access, and ensuring data integrity. A DBA also works closely with developers and system administrators to optimize database performance and support business applications.

Responsibilities of a Database Administrator (DBA)

1. **Database Installation and Configuration**
   a. Install, configure, and upgrade database management systems (DBMS).
   b. Ensure proper setup of database servers.
2. **Database Design and Implementation**
   a. Create and maintain database structures such as tables, indexes, and views.
   b. Ensure efficient database design.
3. **Security Management**
   a. Create user accounts and assign permissions.
   b. Protect databases from unauthorized access and cyber threats.
4. **Backup and Recovery**
   a. Perform regular database backups.
   b. Restore data in case of system failure or data loss.
5. **Performance Monitoring and Tuning**
   a. Monitor database performance.

b. Optimize queries, indexes, and storage for better efficiency.

6. **Data Integrity and Consistency**
   a. Ensure data is accurate, consistent, and reliable.
   b. Enforce integrity constraints and validation rules.
7. **Database Maintenance**
   a. Perform routine maintenance tasks.
   b. Update software patches and fix database issues.
8. **Disaster Recovery Planning**
   a. Develop recovery strategies to minimize downtime.
   b. Ensure business continuity during system failures.
9. **Storage Management**
   a. Manage database storage space.
   b. Allocate additional storage when required.
10. **Documentation and Support**
    a. Maintain database documentation.
    b. Provide technical support to users and developers.

Conclusion

A **Database Administrator (DBA)** plays a vital role in ensuring that database systems are secure, reliable, and efficient. By managing installation, security, backup, recovery, performance optimization, and maintenance, a DBA ensures that organizational data is always available, accurate, and protected, enabling smooth and uninterrupted business operations.